# PDF Materials: Data Cleaning Basics

Sample PDF reference | Data Science | MBU DS LAB

## Why clean data?

Real-world datasets may contain missing values, duplicate rows, inconsistent formats, and outliers. Cleaning helps make analysis more reliable.

## Common checks

- Inspect dataset shape and column names.
- Check missing values and duplicates.
- Standardize date, category, and unit formats.
- Validate ranges and data types.
- Document each change.

## Simple example

If a dataset contains the same student record twice, identify the duplicate and determine whether it should be removed. Never delete records without understanding why they are duplicated.

## Practice task

Download or create a small CSV dataset and write down three quality issues before cleaning it.

*Sample educational material for the MBU DS LAB Resources page. Verify course-specific requirements with your faculty.*